

Unsupervised Machine Learning on Wine Quality Data Clustering, Dimensionality Reduction, and Anomaly Detection Using the UCI Wine Quality Dataset

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Project Objective

What Is Unsupervised Learning?	Goal:	Objective	Business Relevance	Why it matters
• Learns patterns without known target labels • Relies on similarity, distance, and structure • Identifies hidden relationships in data	• Identify natural structure within wine chemistry data without using target labels.	• Identify the following: • Natural wine populations • Hidden subgroup structure • Low-dimensional representations • Rare and unusual wine observations	• Understanding wine chemistry structure can support: • Quality analysis • Product segmentation • Process monitoring • Rare profile detection	• Understanding hidden wine structure can support quality analysis, anomaly detection, and future predictive modeling.

Dataset Description

UCI Wine Quality Dataset

- Physicochemical measurements collected from Portuguese Vinho Verde red and white wines.

Dataset Size

- 5,320 observations
- 11 chemistry features
- 2 wine types
- Quality scores ranging from 3 to 9

Wine Types

- Red Wine
- White Wine

Variable	Description	Was this Feature used for Modeling?
Fixed Acidity	Tartaric acid content	Yes
Volatile Acidity	Acetic acid concentration	Yes
Citric Acid	Citric acid level	Yes
Residual Sugar	Remaining sugar	Yes
Chlorides	Salt concentration	Yes
Free Sulfur Dioxide	SO_2 level	Yes
Total Sulfur Dioxide	Total SO_2	Yes
Density	Wine density	Yes
pH	Acidity measure	Yes
Sulphates	Sulfate concentration	Yes
Alcohol	Alcohol percentage	Yes
Quality	Wine quality score	Used for interpretation and evaluation
Type	Red or White	Used for interpretation and evaluation

Brief View of Wine Data

Variable	Mean	Range
Fixed Acidity (g/dm^3)	7.22	3.8-15.9
Volatile Acidity (g/dm^3)	0.34	0.08-1.58
Residual Sugar (g/dm^3)	5.05	0.6-65.8
Total Sulfur Dioxide (mg/dm^3)	114.11	6-440
Density (g/cm^3)	0.9945	0.987-1.039
pH	3.22	2.72-4.01
Alcohol (% Vol.)	10.55	8.0-14.9

Metric	Value
Observations	5,320
Features	11 Chemistry Variables
Wine Types	Red & White
Quality Range	3-9
Avg Alcohol	10.55%
Avg Residual Sugar	5.05 g/dm^3
Avg pH	3.22
Avg Total Sulfur Dioxide	114.11 mg/dm^3
Metric	Value

Wine Quality Distribution by Wine Type

Key Findings

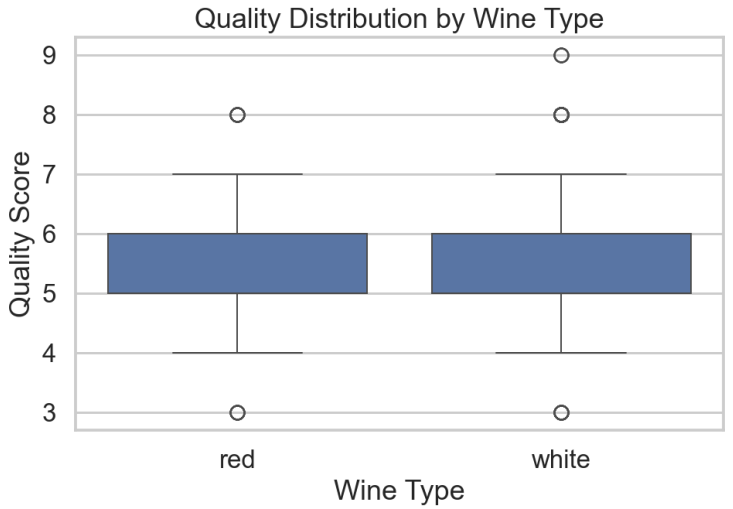
- Red and white wines exhibit very similar quality distributions.
- Both wine types have a median quality score of approximately 6.
- Most wines fall within the quality range of 5 to 6.
- White wines contain substantially more observations than red wines.
- High-quality and low-quality wines are relatively uncommon in both populations.

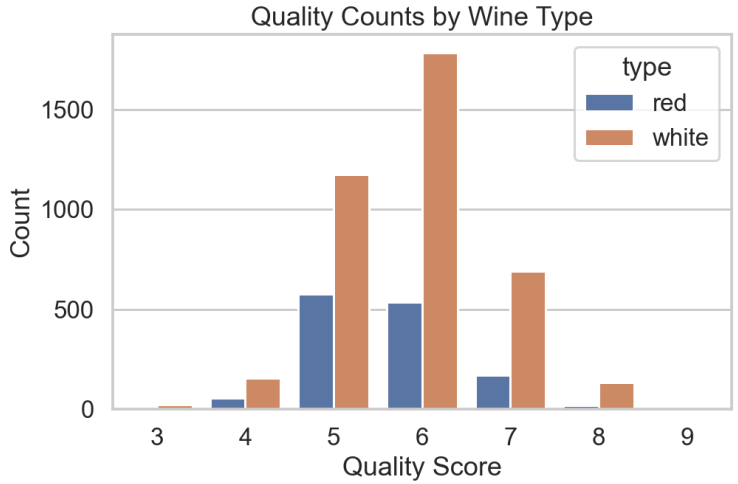
Interpretation

- The similarity in quality distributions suggests that wine type alone does not explain quality differences.
- Quality appears to vary continuously rather than forming distinct groups.
- This observation provides an early indication that quality may not emerge as a dominant structure during clustering or dimensionality-reduction analysis.

Takeaway

- Although red and white wines differ chemically, they display remarkably similar quality distributions. This suggests that factors beyond wine type influence perceived wine quality.





Data Preparation & Analysis Approach



Data Preparation

The original red and white wine datasets were combined into a single analysis dataset and prepared using a consistent preprocessing workflow.



Preparation Steps

Combined red and white wine datasets

Added wine type indicator

Removed duplicate observations

Standardized chemistry features

Prepared feature matrix for unsupervised learning



Why This Matters

Proper preprocessing ensures that all wine chemistry variables contribute fairly to distance calculations, clustering behavior, dimensionality-reduction embeddings, and anomaly-detection results.

Clustering Explained

Why Clustering?

Clustering groups observations with similar characteristics together without using pre-defined labels.

The goal is to determine whether natural wine populations exist based solely on chemistry measurements.

Why Compare Multiple Methods?

Different clustering algorithms make different assumptions about the data.

By comparing multiple approaches, we can determine whether discovered wine populations represent genuine structure rather than artifacts of a single algorithm.

Model Evaluation Criteria:

Silhouette Score

Davies-Bouldin Index

Cluster Interpretability

Visualization Quality

Stability Across Solutions

Clustering Methods

K-Means

- Centroid-based clustering
- Assigns wines to the nearest cluster center
- Works best with compact, spherical groups
- Primary Strength was a fast baseline clustering

Gaussian Mixture Models (GMM)

- Probabilistic clustering
- Allows observations to belong to multiple clusters partially
- Useful when cluster boundaries overlap
- Primary Strength was probabilistic cluster membership

Agglomerative Clustering

- Hierarchical clustering approach
- Builds clusters through progressive merging
- Reveals nested relationships between wines
- Primary Strength was hierarchical structure

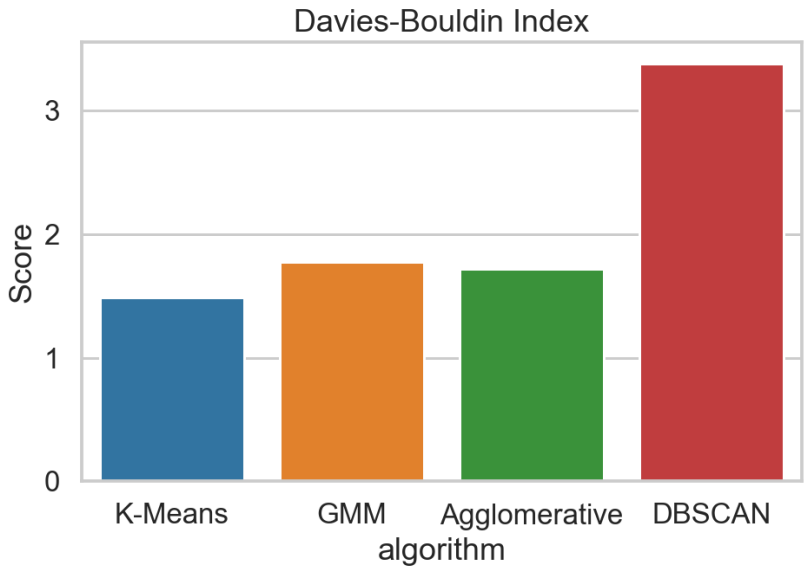
DBSCAN

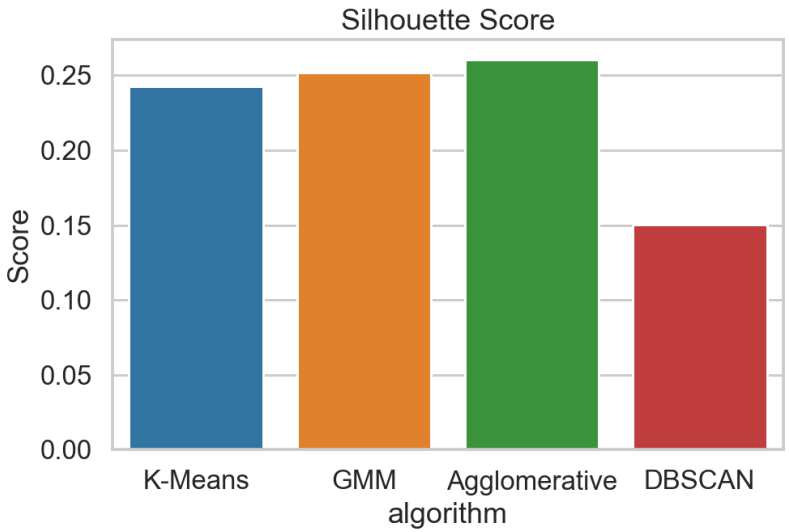
- Density-based clustering
- Detects irregularly shaped clusters
- Automatically identifies noise and outliers
- Primary Strength was density based clustering

Clustering Results & Model Comparison

Key Findings

- Agglomerative Clustering achieved the highest Silhouette Score (0.26).
- K-Means achieved the lowest Davies-Bouldin Index.
- DBSCAN produced the weakest clustering performance.
- Ward Linkage Agglomerative Clustering was selected as the preferred clustering approach.
- K-Means achieved the lowest Davies-Bouldin Index (1.48).
- GMM produced similar performance to K-Means and Agglomerative.
- DBSCAN produced substantially weaker cluster quality across both metrics.





Clustering Results

Clustering Results	**The best clustering solution:** Ward Linkage Agglomerative Clustering	Key Findings	Major Insight	Conclusion
• Four clustering approaches were evaluated to determine whether natural wine populations can be inferred from chemical measurements alone.	• Produced clear wine population separation • Revealed meaningful hierarchical structure • Identified potential subgroup relationships • Generated interpretable dendrograms • Consistent with dimensionality-reduction findings	• Red and white wines formed distinct populations across multiple clustering methods. • Cluster quality consistently improved when hierarchical structure was considered. • Hierarchical clustering revealed additional subgroup structure within both wine populations. • Wine quality scores remained distributed across clusters and did not form independent clusters.	• The strongest clustering signal in the dataset was Wine type. • Strongest cluster was Red Wines vs. White Wines • Additional evidence suggested nested subgroup structure within both populations.	• Multiple clustering methods independently identified wine type as the dominant source of variation within the dataset, while hierarchical clustering provided the most interpretable representation of wine population structure.

Dimensionality Reduction Methods

Objective

- Determine which dimensionality-reduction technique provides the most informative representation of wine chemistry while preserving meaningful structure.

Why Dimensionality Reduction?

- Wine chemistry contains multiple correlated variables that can be difficult to visualize and interpret simultaneously.
- Dimensionality reduction transforms high-dimensional chemistry data into a smaller set of variables while preserving as much meaningful structure as possible.
- This method helps:
  - Visualize wine populations
  - Reveal hidden structure
  - Identify nonlinear relationships
  - Reduce dimensional complexity
  - Improve interpretation

Evaluation Goals

- Each method was evaluated on:
- Interpretability
- Visual separation
- Structure preservation
- Nonlinear relationship discovery
- Chemistry factor interpretation

Key Question

- Which dimensionality-reduction technique provides the most informative low-dimensional representation of the Wine Quality dataset?

Methods Evaluated

Principal Component Analysis (PCA)

- **What is it?**
  - A linear dimensionality-reduction technique that identifies directions of maximum variance within the data.
- **Why use it?**
  - Preserves variance
  - Produces interpretable loadings
  - Establishes a baseline representation

Kernel PCA

- **What is it?**
  - A nonlinear extension of PCA that uses kernel functions to reveal curved and complex relationships.
- **Why use it?**
  - Captures nonlinear structure
  - Improves separation when PCA is limited
  - Reveals hidden geometry in feature space

t-SNE

- **What is it?**
  - A nonlinear embedding method that focuses on preserving local neighborhood relationships.
- **Why use it?**
  - Excellent for visualization
  - Reveals cluster separation
  - Highlights local population structure

UMAP

- **What is it?**
  - A nonlinear manifold-learning algorithm that preserves both local and global relationships.
- **Why use it?**
  - Strong cluster separation
  - Maintains broader dataset structure
  - Generates highly interpretable visualizations

Non-Negative Matrix Factorization (NMF)

- **What is it?**
  - A factorization technique that represents data as additive non-negative components.
- **Why use it?**
  - Produces interpretable latent factors
  - Identifies chemistry building blocks
  - Explains how wine characteristics combine

Methods Major Findings

PCA

- Explained nearly 50% of variance using two components
- Clearly separated red and white wines
- Provided interpretable feature loadings
- Best method for Variance Explanation
- Primary strength is Variance Explanation

Kernel PCA

- Revealed nonlinear chemical structure
- Improved separation relative to standard PCA
- Best method for Nonlinear Structure Discovery
- Primary strength: Nonlinear Structure Discovery

t-SNE

- Produced outstanding local cluster separation
- Revealed highly distinct wine populations
- Best method for Local Cluster Separation
- Primary strength is nonlinear structure discovery

UMAP

- Produced the strongest overall embedding
- Balanced separation and structure preservation
- Generated the clearest visualization
- Best method for overall visualization
- Primary strength was local and global structure preservation

NMF

- Identified interpretable chemistry components
- Revealed latent additive wine characteristics
- Best Method for feature interpretation
- Primary strength was latent chemistry interpretation

Findings and Conclusion

Most Important Finding:

- Wine Type was the Dominant Source of Variation,while Quality Scores remained Highly Overlapping
- All dimensionality-reduction methods consistently identified wine type as the strongest source of structure within the dataset.

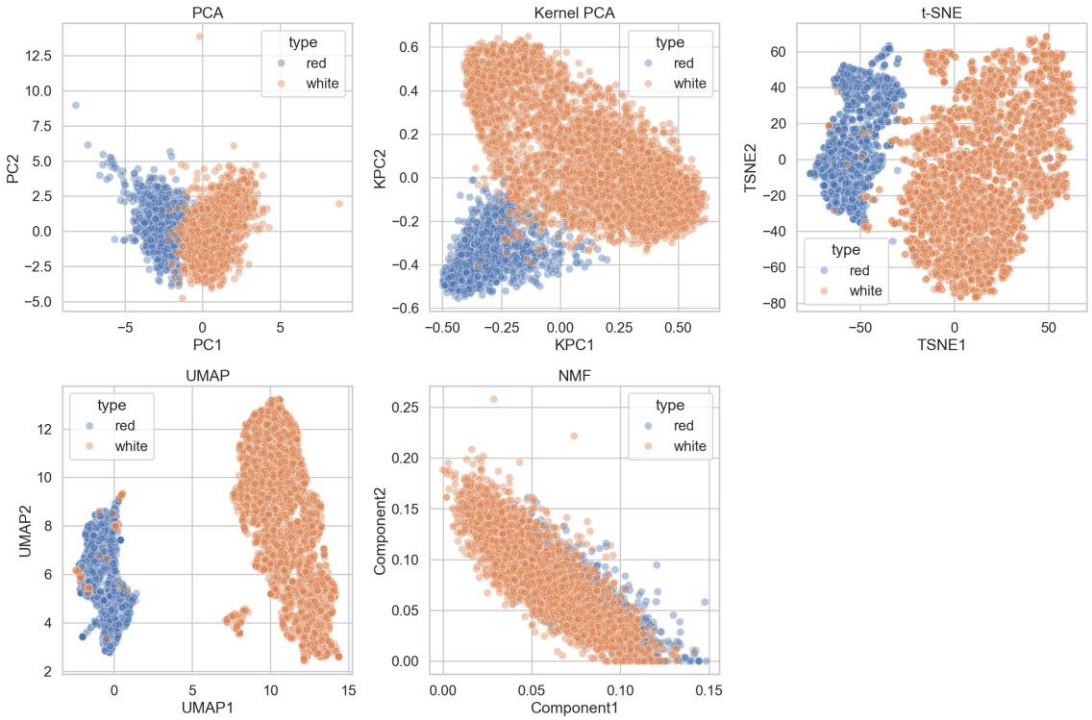
Conclusion

- Multiple dimensionality-reduction techniques independently arrived at the same conclusion:
- Red and white wines form distinct populations.
- Quality scores do not form clearly separated groups.
- UMAP provided the strongest overall visualization.
- NMF delivered the strongest chemistry interpretation.

Dimensionality Reduction Comparison

Key Findings

- All dimensionality-reduction methods identified **wine type** as the dominant source of variation.
- Red and white wines formed distinct populations across multiple embeddings.
- Nonlinear methods produced stronger separation than linear PCA.
- Wine quality remained highly overlapping and did not form clear low-dimensional groups.
- Multiple methods suggested evidence of internal subgroup structure.



Anomaly Detection Methods

Why Anomaly Detection?

- Not all wines fit naturally into clusters or populations.
- Anomaly detection identifies wines whose chemistry differs significantly from the rest of the dataset.
- These observations may represent:
  - Rare chemical profiles
  - Production inconsistencies
  - Extreme wine styles
  - Observations that influence clustering structure

Isolation Forest

- **What is it?**
  - An unsupervised anomaly-detection algorithm that isolates observations using random feature splits.
- **How does it work?**
  - Observations that require fewer splits to isolate are considered more unusual.
- **What does it detect?**
  - Global anomalies
  - Chemistry extremes
  - Observations far from the overall population

Local Outlier Factor (LOF)

- **What is it?**
  - A density-based anomaly-detection algorithm that compares each observation to its nearest neighbors.
- **How does it work?**
  - Observations with substantially lower local density than surrounding wines are classified as anomalous.
- **What does it detect?**
  - Local anomalies
  - Sparse neighborhood observations
  - Unusual wines hidden within larger populations

Why Compare Both Methods?

- Both methods define anomalies differently.
- Isolation Forest → Globally Unusual Wines
- LOF → Locally Unusual Wines
- Using both approaches provides a more complete understanding of rare and unusual wine chemistry profiles.

Key Question

- Do globally unusual wines and locally unusual wines represent the same observations?

Anomaly Detection Results

Objective

- Determine whether unusual wine observations exist within the dataset and evaluate how different anomaly-detection techniques characterize rare wine chemistry profiles.

Isolation Forest

- Focused on globally unusual wines
- Identified predominantly red wine anomalies
- Anomalies Detected
  - Total:266
  - Red:164
  - White: 102
- Highlighted chemistry extremes
- Most anomalies occurred along outer boundaries of feature space

Local Outlier Factor (LOF)

- Focused on locally unusual wines
- Identified predominantly white wine anomalies
- Anomalies Detected
  - Total:266
  - Red: 87
  - White: 179
- Detected sparse neighborhood behavior
- Found unusual wines throughout feature space

Most Important Insight

- Global Anomalies ≠ Local Anomalies
- Only 36.09% of anomalies were identified by both methods.
- This suggests that rare wine behavior exists at multiple scales and cannot be fully characterized using a single anomaly-detection technique.

Conclusion

- Isolation Forest and LOF revealed complementary aspects of anomaly structure within the Wine Quality dataset.
- Red wines were more likely to exhibit globally extreme chemistry profiles, while white wines were more likely to exhibit locally unusual chemistry patterns.

Key Findings & Insights

Wine Type Is the Dominant Source of Variation

- Across clustering, dimensionality reduction, and anomaly-detection methods, red and white wines consistently formed distinct populations.
- Evidence was observed in:
  - Clustering results
  - Hierarchical clustering
  - PCA
  - Kernel PCA
  - t-SNE
  - UMAP
  - NMF

Wine Quality Does Not Form Natural Groups

- Wine quality scores remained highly overlapping across:
- Clusters
- PCA space
- Nonlinear embeddings
- Latent-factor representations
- This suggests that wine quality behaves more like a continuous spectrum than a set of distinct chemistry groups.

Additional Subgroup Structure Exists

- Ward Linkage, Kernel PCA, t-SNE, and UMAP all suggested the presence of meaningful subpopulations within both red and white wines.
- The data contains more structure than a simple:
  - Red Wine vs. White Wine separation

UMAP Produced the Strongest Visualization

- Provided the clearest representation of the dataset
- Among all dimensionality-reduction methods:
  - Best separation
  - Best structure preservation
  - Most interpretable embedding

Different Anomaly Methods Reveal Different Behaviors

- Only **36.09%** of anomalies were shared between methods.
- This indicates that anomaly behavior depends strongly on how unusual observations are defined.

Overall Project Insight

- Multiple independent methods arrived at the same conclusion:
  - Wine Type drives the majority of structure within the Wine Quality dataset.

Final Recommendations

Recommended Analytical Workflow

- Ward Clustering
  ↓
  UMAP Visualization
  ↓
  NMF Interpretation
  ↓
  Isolation Forest + LOF
- This combination provides the strongest balance between:
  - Structure discovery
  - Interpretability
  - Visualization quality
  - Outlier identification

Ward Linkage

- Produced the most interpretable clustering structure
- Revealed hierarchical relationships between wines
- Identified meaningful subgroup structure

UMAP

- Generated the clearest low-dimensional representation
- Preserved local and global structure
- Consistently separated wine populations

NMF

- Revealed interpretable latent chemistry components
- Explained how wine characteristics combine

Isolation Forest & LOF

- Identified complementary anomaly types
- Captured both global and local outlier behavior

Practical Takeaway

- No single unsupervised-learning algorithm was optimal for every task.
- The strongest results were obtained by combining multiple complementary techniques.
- Using clustering, dimensionality reduction, and anomaly detection together provided a much more complete understanding of wine chemistry structure than any method alone.

Goals and Recommended Methods

Analysis Goal	Recommended Method
Overall Clustering	Ward Linkage Agglomerative Clustering
Hierarchical Structure Discovery	Ward Linkage
Variance Explanation	PCA
Nonlinear Structure Discovery	Kernel PCA
Local Cluster Separation	t-SNE
Overall Visualization	UMAP
Chemistry Interpretation	NMF
Global Anomaly Detection	Isolation Forest
Local Anomaly Detection	LOF

Dataset Limitations

Unsupervised Learning Challenges

- Unlike supervised learning, there is no known "correct" answer for clusters, latent factors, or anomalies.
- As a result, many findings require interpretation rather than direct validation.

Dataset Feature Limitations

- The dataset contains only physicochemical measurements and does not include:
- Grape variety
- Region of origin
- Production methods
- Storage conditions
- Expert tasting notes
- These factors may provide additional insight into wine behavior.

Quality Scores Remain Subjective

- Wine quality scores are based on sensory evaluations and may vary across evaluators.
- As a result:
  - Quality may not align perfectly with chemistry
  - Quality groups may naturally overlap

Method Limitations

Clustering Method Limitations

- Cluster interpretation is subjective
- Different algorithms may produce different group structures
- No ground-truth cluster labels exist

Dimensionality Reduction Method Limitations

- PCA may overlook nonlinear structure
- t-SNE and UMAP can distort distances during projection
- Visual separation does not always imply true cluster structure

Anomaly Detection Method Limitations

- Isolation Forest and LOF use different definitions of anomalies
  - Results depend on algorithm assumptions
  - Rare observations are not always errors or defects

Key Limitation

- Unsupervised methods identify structure, but they cannot directly explain why that structure exists.
- Domain knowledge is still required to interpret results correctly.

Next Steps

Additional Clustering Approaches

- Evaluate more advanced clustering techniques:
- HDBSCAN
- Spectral Clustering
- Affinity Propagation
- These methods may reveal additional subgroup structure not detected by current models.

Enhance Distance & Similarity Analysis

- Investigate alternative distance metrics:
- Manhattan Distance
- Cosine Similarity
- Mahalanobis Distance
- Different distance definitions may alter cluster structure and anomaly detection performance.\

Improve Anomaly Analysis

- Evaluate the impact of anomalies on clustering results by:
- Removing detected anomalies
- Re-running clustering methods
- Comparing cluster stability before and after removal
- This would help determine how strongly rare wines influence discovered populations.

Add Cluster Interpretation

- Develop a hybrid supervised-unsupervised workflow:
- Cluster Assignment → Classification Model → Feature Importance / SHAP
- This would provide direct explanations for cluster membership and improve interpretability.

Extend Dimensionality Reduction

- Future analysis could explore:
- Three-dimensional embeddings
- Autoencoders
- Variational Autoencoders (VAE)
- Deep representation learning
- These methods may reveal additional nonlinear structure.

Long-Term Goal

- Develop an integrated exploration framework that combines:
  - Clustering
  - Dimensionality Reduction
  - Anomaly Detection
- Explainable AI to support deeper understanding of wine chemistry and population structure.

Project Conclusion

Project Objective Achieved

- This project successfully applied clustering, dimensionality reduction, and anomaly-detection techniques to uncover hidden structure within the Wine Quality dataset.

Major Conclusions

- Wine Type Drives Dataset Structure – Red and white wines consistently formed distinct populations across nearly all unsupervised learning methods evaluated.
- Wine Quality Remains Highly Overlapping– Quality scores did not form clearly separated chemistry groups and behaved more like a continuous spectrum than distinct categories.
- Additional Subgroups Exist – Multiple methods revealed evidence of meaningful subgroup structure within both red and white wine populations.
- No Single Method Was Best at Everything
- Multiple Methods Reinforced the Same Findings – One of the strongest outcomes of the project was the consistency of results across independent techniques.
- Clustering, dimensionality reduction, and anomaly detection all pointed toward the same fundamental structure within the dataset.

Final Takeaway

- Wine Type → Primary Structure
- Subgroups → Secondary Structure
- Quality → Highly Overlapping
- Anomalies → Method Dependent
- Together, these findings provide a comprehensive understanding of wine chemistry structure and demonstrate the value of combining multiple unsupervised-learning techniques when exploring complex datasets.